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 Studierichting: Informatica
 Oefeningenreeks 3A nr. 36.16

$$\begin{aligned}
 f(x) &= \ln(x^3 + \sqrt{x^6 + 1}) \\
 f(x) &= \ln(g(x)) \text{ met } g(x) = x^3 + \sqrt{x^6 + 1} \\
 f'(x) &= \frac{g'(x)}{g(x)} \\
 &= \frac{g'(x)}{x^3 + \sqrt{x^6 + 1}} \\
 &= \frac{3x^2 + (\sqrt{x^6 + 1})'}{x^3 + \sqrt{x^6 + 1}} \\
 &= \frac{3x^2 + \frac{1}{2\sqrt{x^6 + 1}}6x^5}{x^3 + \sqrt{x^6 + 1}} \\
 &= \frac{3x^2 + \frac{6x^5}{2\sqrt{x^6 + 1}}}{x^3 + \sqrt{x^6 + 1}} \\
 &= \frac{3x^2 + \frac{3x^5}{\sqrt{x^6 + 1}}}{x^3 + \sqrt{x^6 + 1}} \\
 &= \frac{3x^2 \left(1 + \frac{x^3}{\sqrt{x^6 + 1}}\right)}{x^3 + \sqrt{x^6 + 1}} \\
 &= \frac{3x^2 \left(\frac{\sqrt{x^6 + 1}}{\sqrt{x^6 + 1}} + \frac{x^3}{\sqrt{x^6 + 1}}\right)}{x^3 + \sqrt{x^6 + 1}} \\
 &= \frac{3x^2 \left(\frac{x^3 + \sqrt{x^6 + 1}}{\sqrt{x^6 + 1}}\right)}{x^3 + \sqrt{x^6 + 1}} \\
 &= 3x^2 \left(\frac{x^3 + \sqrt{x^6 + 1}}{\sqrt{x^6 + 1}}\right) \frac{1}{x^3 + \sqrt{x^6 + 1}} \\
 &= \frac{3x^2}{\sqrt{x^6 + 1}}
 \end{aligned}$$

References